



# Depth-Adaptive Discriminant Projection with Optimal Transport

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**Abstract.** Discriminant projection is a key technique for dimensionality reduction, especially in many classification tasks of high-dimensional, small-sample datasets. In this paper, we propose a Depth-Adaptive Discriminant Projection (DADP) method to improve the discriminability of the projection subspace. First, we propose a novel single-layer discriminant projection by adjusting projected points with nonlinear transformation. Second, the single-layer structure is extended to multiple-layer ones according to the proposed Adaptive Depth Determination Criterion (ADDC), naturally constructing a deep projection model with adaptive depth, which makes DADP allow for the avoidance of over-fitting problem for small-scale datasets. Furthermore, the regularized optimal transport (OT) is introduced to learn the optimal weights for pairwise data, which balances the local and global information incorporated in scatter estimation. Experimental results on several high-dimensional, small-scale datasets show the effectiveness of our algorithm both in terms of visualization and classification prediction.

**Keywords:** Dimensionality reduction · Optimal transport · Adaptive depth

## 1 Introduction

Dimensionality reduction (DR) as a crucial technique for analyzing real-world data, aims at learning a more meaningful and compact low-dimensional representation of high-dimensional data. One major family of DR is discriminant projection, the goal of which is to learn a subspace that preserves class separability for the tasks such as clustering and classification. When label information is available, various supervised discriminant projection techniques [3, 4, 7, 10] have been explored to learn a low-dimensional subspace which both preserve intrinsic structure and discriminative information of input data. The main characteristic of discriminant subspace is that samples from different classes are as separate

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as possible while samples from the same class are close to each other. Linear Discriminant Analysis (LDA), a classic linear dimensionality reduction method, seeks to find such projection subspace by maximizing the ratio of inter-class and intra-class scatter. However, like most linear models, LDA is hard to uncover the structure of nonlinear data that are very universal in realistic applications. Thus, a large number of nonlinear variants have been developed for addressing this problem.

With the assumption that real-world data could be linearly separable in some space with much higher dimensionality, kernel based techniques, such as Kernel Fisher Discriminant Analysis (KFDA) [5, 9] implicitly project observed data into a high-dimensional feature space via a nonlinear mapping determined by kernel function. However, choosing an appropriate kernel for a specific dataset is challenging due to the lack of prior knowledge, which limits the application of kernel based techniques. In contrast, deep architecture based methods directly learn an explicit nonlinear mapping via deep neural network instead of manually selecting a kernel function. GerDA [6] attempts to learn a linearly separable latent representation by fine-tuning a pre-trained restricted Boltzmann machines on a linearly discriminant criterion. For simplicity, DeepLDA [2, 8] directly replaces cross entropy loss with Fisher discriminant loss and optimizes several smallest eigenvalues for increasing discriminative variance in all projection directions. However, an optimal nonlinear subspace cannot be learned by the network with an inappropriate architecture. Apart from that, except for the last layer corresponding to a linearly separable subspace, each of the former layers lacks interpretability from the perspective of discriminability. Other problems like the need of large samples to avoid over-fitting and the extreme sensitivity to hyper-parameter setting, also severely limit the application of such kinds of methods in real-world problems [11].

Motivated by the limitations of existing methods, we propose a general discriminant subspace learning algorithm called Depth-adaptive Discriminant Projection (DADP), which shares a similar architecture of DNN. Compared with previous works based on DNN, DADP endows the discriminability to each single-layer nonlinear mapping. Moreover, to enhance the discriminative capacity layer-by-layer, the Adaptive Depth Determination Criterion (ADDC) is proposed to determine the depth of our deep architecture. The key idea is to seek a projection space which has a better discriminative performance than that of previous layer. The layer-by-layer projection process would not terminate until there is no more a subspace with better separability. Therefore, the proposed DADP method can adaptively determine the depth for a given dataset.

The main contributions of this paper are summarized in the following three points,

- 1) Adaptive depth: With the help of ADDC, DADP can adaptively determine the depth for a given real-world datasets. Meanwhile, DADP can avoid the problem of over-fitting for small sample size cases.
- 2) Optimal projection matrix: DADP selects optimal weighting plans for constructing the inter-class and intra-class scatter matrix respectively via

smoothed optimal transport. Our approach adaptively balances the local and global data correlation incorporated in scatter estimation, which further improves the discriminability of subspace for given datasets.

- 3) Unrestricted projection dimension: In contrast to LDA, projection dimension of DADP is not bounded by the number of classes. Each layer mapping can freely project data into any lower-dimensional subspace.

## 2 Preliminary

### 2.1 Notations

Let  $\langle A, B \rangle := \text{tr}(A^T B)$  denote the Frobenius dot-product of two matrix. Besides, we note the simplex as  $\sum_d := \{x \in \mathbb{R}_+^d : x^T \mathbf{1}_d = 1\}$ , where  $\mathbf{1}_d$  is the  $d$  dimensional vector of ones. For two probability vector  $r$  and  $c$  in the simplex  $\sum_d$ , we write  $U(r, c)$  be the transport polytope of  $r$  and  $c$ , namely the polyhedral set of  $d \times d$  matrices, which satisfies  $U(r, c) := \{T \in R_+^{d \times d} | T\mathbf{1}_d = r, T^T \mathbf{1}_d = c\}$ .

### 2.2 Regularized Optimal Transport

Choosing a suitable metric to compare probability distributions is crucial to many machine learning tasks. Among traditional divergences, the distance defined by optimal transport is capable of capturing geometrical information when the probability space is a metric space. The distance, known as Wasserstein distance or Earth Mover's distance, is solved by linear programming with complexity at least in  $O(d^3 \log(d))$ . To facilitate the applicability of OT in large-scale data, an entropic regularization is introduced to obtain a smoothed version of OT. More importantly, regularized OT can be solved with Sinkhorn's matrix scaling algorithm, which can be computed several orders of magnitude faster [1].

**Definition 2.** (Regularized Optimal Transport) Given a  $d \times d$  cost matrix  $M$ , the cost of mapping  $r$  to  $c$  using a transport matrix (or a joint probability)  $T$  can be quantified as  $\langle T, M \rangle$ , the regularized optimal transport is defined as,

$$d_M^\lambda(r, c) := \langle T^\lambda, M \rangle \quad (1)$$

$$T^\lambda = \arg \min_{T \in U(r, c)} \langle T, M \rangle - \frac{1}{\lambda} H(T) \quad (2)$$

where  $H(T) = -\sum_{i=1}^d \sum_{j=1}^d T_{ij} \log T_{ij}$  is the entropy of  $P$ , and  $\lambda > 0$  is entropic regularization coefficient.

It is noteworthy that the optimal transport  $T^*$  is a sparse  $d \times d$  matrix with at most  $2d - 1$  non-zero elements [1]. In our work, sparsity of  $T$  means those sample points merely correlate with their nearest neighbors and global information of projected data has not been sufficiently considered. Hence, smooth regularizers (entropic regularization e.g.) are necessary to take both local and global interactions into account.

### 3 The Proposed Method

#### 3.1 Problem Statements

In many real-world applications, the volume of available data is ordinarily limited while the dimensionality is very high. Hence, we consider the problem of learning a discriminative subspace for small-scale high-dimensional data. Given a dataset  $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_m = \mathbf{X} \in \mathbb{R}^{d \times m}$ , which contains  $m$  samples from  $C$  different classes  $c = \{1, 2, \dots, C\}$ . We define set  $\zeta = \{(c_1, c_2) | c_1 \neq c_2\}$ , denoting the set of all pairs of different categories. The goal is to find a target projected subspace which preserves class discriminability.

#### 3.2 Single-Layer Discriminant Projection

- 1) *Characterize the inter-class Scatter.* Given that the optimal linear projection matrix is  $W \in \mathbb{R}^{d \times l}$ ,  $l \leq d$ . Regarding the determination of an optimal projection dimensionality  $l$ , we will have a detail discussion in later section. The distance between pairs of data from different classes is  $\|W^T \mathbf{x}_i - W^T \mathbf{x}'_j\|$ . By tuning the regularization term  $\lambda$ , we attempt to learn the appropriate weights for all pairs from  $\zeta$ . The weighted sum of the squared distances of point pairs in  $\zeta$  can be written as:

$$\begin{aligned} d_b &= \sum_{i,j \in \zeta} \text{tr} T_{i,j} (W^T \mathbf{x}_i - W^T \mathbf{x}'_j) (W^T \mathbf{x}_i - W^T \mathbf{x}'_j)^T \\ &= \text{tr} W^T \sum_i \sum_j T_{i,j} (\mathbf{x}_i - \mathbf{x}'_j) (\mathbf{x}_i - \mathbf{x}'_j)^T W \\ &= \text{tr} W^T S_b W \end{aligned} \quad (3)$$

where  $S_b = \sum_{i,j \in \zeta} T_{i,j} (\mathbf{x}_i - \mathbf{x}'_j) (\mathbf{x}_i - \mathbf{x}'_j)^T$  is the inter-class scatter matrix,

$T_{i,j}$  is the transportation weight that is obtained by optimizing a regularized optimal transport problem Eq. (2). The cost matrix  $M$  is the pairwise squared Euclidean distance  $M_{i,j} = \|\mathbf{x}_i - \mathbf{x}'_j\|^2 \in \mathbb{R}^{n \times m}$ , where  $\mathbf{x}_i \in c_1, \mathbf{x}'_j \in c_2$ ,  $n$  and  $m$  are the number of samples in class  $c_1$  and  $c_2$  respectively. Note that,  $\mathbf{x}_i$  and  $\mathbf{x}'_j$  are feature vectors passed from the last layer. Here, according to Eq. (2), when cost matrix  $M$  is fixed,  $T_{i,j}(\lambda_1)$  can be viewed as a function of  $\lambda_1$ . Note that, when  $\lambda_1$  is large enough, the impact of entropic regularization is so small that the problem boils down to the origin optimal transport. Even in special cases where two classes have equal number of points, the problem has become an optimal matching problem in which each point only is linked to one another. Hence, it is crucial to set an appropriate  $\lambda_1$  for selecting the effective data pairs.

- 2) *Characterize the intra-class Scatter.* Similarly, we have weighted sum of squared distances of pairwise points from the same class  $c = \{1, 2, \dots, C\}$ ,

$$d_w = \text{tr} (W^T S_w W) \quad (4)$$

where  $S_w = \sum_{i,j \in c} T'_{i,j} (x_i - x_j)(x_i - x_j)^T$  is the intra-class scatter matrix.

$T'_{i,j}(\lambda_2)$  is also a function of  $\lambda_2$ . Similar to the discussion above, when  $\lambda_2$  is small enough, regularization term plays the dominant role in the optimization problem. When  $\lambda_2 \rightarrow 0$ , the transport matrix  $T := rc^T$  which indicates two distributions are independent to each other and all coupling are used to construct the intra-class scatter. Conversely, when  $\lambda_2 \rightarrow \infty$ , optimal transport cost from an empirical distribution to itself is 0. From the point of the correlation of points, one point only interacts with itself, which is incapable of capturing the global structure of data from the same class. Therefore it is necessary to select a relatively smaller value for  $\lambda_2$  ( $\lambda_2 \ll \lambda_1$ ) to obtain a smoother transport matrix  $T'$ .

**Optimal Projection Matrix Selection.** We cast optimal projection matrix selection as an optimization problem with the objective of maximizing the trace of inter-class scatter of projected data while minimizing the trace of intra-class scatter of projected data. The objective function is formulated as,

$$J(W) = \max \frac{\text{tr}(W^T S_b W)}{\text{tr}(W^T S_w W)} \quad (5)$$

$$\text{s.t. } W^T W = I$$

where  $W \in R^{d \times l}$ ,  $I$  is  $d \times d$  identity matrix. Note that the formulation of  $J(W)$  is in form of Raleigh quotients, the solution of optimal  $W^*$  can be obtained by solving the general eigenvalue problem,  $S_b W = \alpha S_w W$  where  $W = (e_1, e_2, \dots, e_n)$ ,  $e_i$  is eigenvector of  $S_w^{-1} S_b$  and the corresponding eigenvalue is  $\alpha_i$ . To avoid the matrix singularity problem, we adjust the intra-class scatter matrix by adding a multiple of identity matrix  $I$ , as follows,  $(S_w + \gamma I)^{-1} S_b e = \alpha e$ .

**Nonlinear Activation.** Inspired by the single-layer nonlinear transformation of deep neural network, we adopt similar nonlinear activation approach via *sigmoid*( $\cdot$ ) function. Besides, we adjust linearly projected data by adding an extra bias vector  $b$  before nonlinear activation. To obtain a better nonlinear projection subspace, we build the objective function as follows,

$$\min_b \sum_{i=1}^C \sum_{y_j \in C_i} \|\delta(y_j + b) - m_i\|^2 \quad (6)$$

where  $C$  is the number of total classes, and  $\delta(y) = (1 + \exp(y))^{-1}$ ,  $m_i = \frac{1}{K_i} \sum \delta(y_j + b)$  is the mean vector of  $i$ -th class. As can be seen from Eq. (6), we hope that the distances of point pairs from the same class are as small as possible after nonlinear mapping. In this paper, we find the bias vector using gradient descent. In addition, we pre-processing the input data or latent representation by whitening before nonlinear mapping at each layer.

### 3.3 Multiple-Layer Discriminant Projection

It is well-known that powerful data representation learning ability of deep neural network stems from deep architecture, which motivates us to extend single-layer discriminant projection to multiple-layer ones for boosting the discriminability of embedding subspace. Although deep architecture is effective in realistic tasks, it is still intractable to decide how many layers of network are suitable for a given dataset. Hence, we propose a criterion called Adaptive Depth Determination Criterion (ADDC) to adaptively determine the depth of network and select the dimensionality of projection corresponding to each layer.

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**Algorithm 1.** Framework of depth-adaptive discriminant projection.

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**Input:**

Training set  $S = \{(x_1, y_1), (x_2, y_2) \dots (x_m, y_m)\}$ ,  
 $\lambda_1 \geq 0, \lambda_2 \geq 0, \gamma \geq 0, 0 < \varepsilon < 1$ ;

**Output:**

Adaptive deep discriminant projection model  $f(x)$ ;

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1: Initialize  $p_1 = p_0 - 1, l = 1$ , whitening the training set  $S$ , calculate  $S_w, S_b$  respectively according to Eq. (3), Eq. (4);
2: while 1 do
3:   if  $l = 1$  then
4:     calculate  $W, b$  according to Eq. (5), Eq. (6).
5:     while  $acc(p_1) < \frac{1}{C}$  do
6:        $p_1 = p_1 - 1$ 
7:       update  $W, b$ 
8:     end while
9:      $l = l + 1$ 
10:  end if
11:  update  $S_w, S_b$ , calculate  $p_l$  according to Eq. (8).
12:  update  $W, b$ 
13:  if  $acc(p_l) > acc(p_{l-1})$  then
14:     $l = l + 1$ 
15:  else
16:    break
17:  end if
18: end while

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**Adaptive Depth Determination Criterion.** ADDC determines the projection dimension sequence by two parts, including *initial projection dimension determination* and *subsequent projection dimensions determination*. Here, the quality of the projection subspace is judged by the classification accuracy of a k nearest neighbor algorithm (KNN).

- 1) *Initial Projection Dimension Determination.* Motivated by the idea of weak classifier in the ensemble learning, we propose a general criterion for determining the initial projection dimension  $p_1$  that raw input data are projected

on, the  $p_1$ -dimensional subspace should meet the following conditions: *classification accuracy exceeds random guesses*  $\frac{1}{C}$ . Simultaneously, to increase the depth of whole network as possible, the first projection dimension  $p_1$  should be as large as possible. Overall, the criterion can be formulated as follows,

$$\begin{aligned} & \max p_1 \\ & s.t. p_1 < p_0 \\ & acc(p_1) \geq \frac{1}{C} \end{aligned} \quad (7)$$

$acc(p_1)$  is the prediction accuracy of KNN.  $p_0$  is the origin dimension of input data. It is possible that  $p_1$  is very small if we directly search for a lower dimension  $p_1$  that maximizes classification accuracy. Hence, we only constrain the accuracy greater than  $\frac{1}{C}$  (accuracy of random guesses).

- 2) *Subsequent Projection Dimension Determination*. To achieve a depth-adaptive network structure, we determine the subsequent dimensions based on two conditions: (i) the preserve of projection directions with major discriminative variance. (ii) the improvement of the classification accuracy. For the former condition, we preserve those projection directions which correspond to the larger eigenvalues because each eigenvalue  $\alpha_i^{l-1}$  quantifies the amount of separation in direction of  $e_i$  of the  $l-1$  layer. Here, eigenvalues are in descending order and criterion can be written as,

$$p_l = \arg \min_j \frac{\sum_{i=1}^j \alpha_i^{l-1}}{\sum_{i=1}^{p_{l-1}} \alpha_i^{l-1}} \geq \varepsilon \quad (8)$$

where  $\varepsilon$  is the threshold value set around 95%. For the latter condition, if  $p_l$ -dimensional subspace does not meet  $acc(p_l) > acc(p_{l-1})$ ,  $l \geq 2$ , which means there is no improvement of the performance of the current network and the projection process terminates. Thus, the depth of network can be automatically determined, which make DADP applicable to different scaling of datasets.

## 4 Experiments

As discussed above, shallow models that are linear or nonlinear are hard to uncover essential structure of data with high dimensionality, while deep architecture based models require large volume of data. Hence, in this section, we present the evaluation of the subspace learning capacity of DADP on several small-scale high-dimensional datasets. Experiments are performed on 10 UCI dataset, face datasets, such as AR, ORL and FERET databases and Alzheimer's disease datasets that include 4 types. The performance of discriminative projection of DADP is compared with some classical discriminant analysis methods like LDA, LSDA, some locality preserving methods, such as LFDA, LMNN, and a deep neural network based method DeepLDA. The discriminability of the

embedding is judged by the recognition rate of  $k$  nearest neighbors classifier with Euclidean distance. Note that we have split the datasets 80%-20% for training set and testing set respectively and repeated the random partition process 20 times. In order to obtain a reliable numerical results, we take the average of 20 independent experimental results as the final recording. Considering the potential impact brought by the dimensionality of projection subspace, we roughly initialized the dimensionality  $p$  to  $C - 1$  except for DADP, where  $C$  is the number of classes.

**Table 1.** Average test classification accuracy on 10 UCI dataset. Each row represents a dataset. For each dataset, 20 test runs were performed and the average classification accuracy as well as standard deviation are presented. Besides, the highest accuracy for each dataset is in bold.

|                   | LDA               | LFDA              | LMNN                                | LSDA              | DeepLDA                             | DADP                                |
|-------------------|-------------------|-------------------|-------------------------------------|-------------------|-------------------------------------|-------------------------------------|
| Blood transfusion | .5862 $\pm$ .0078 | .5831 $\pm$ .0084 | .6033 $\pm$ .0088                   | .6012 $\pm$ .0073 | .5823 $\pm$ .0142                   | <b>.7782 <math>\pm</math> .0075</b> |
| Breast cancer     | .9942 $\pm$ .0044 | .9956 $\pm$ .0032 | .9972 $\pm$ .0045                   | .9582 $\pm$ .0052 | .8021 $\pm$ .0153                   | <b>.9973 <math>\pm</math> .0031</b> |
| Dermatology       | .9522 $\pm$ .0042 | .9692 $\pm$ .0031 | <b>.9842 <math>\pm</math> .0048</b> | .9157 $\pm$ .0043 | .9122 $\pm$ .0084                   | .9731 $\pm$ .0042                   |
| Diabetes          | .5562 $\pm$ .0076 | .7592 $\pm$ .0074 | .7222 $\pm$ .0086                   | .7549 $\pm$ .0076 | .7533 $\pm$ .0094                   | <b>.8572 <math>\pm</math> .0045</b> |
| Ionosphere        | .9322 $\pm$ .0042 | .9476 $\pm$ .0056 | .9227 $\pm$ .0048                   | .9623 $\pm$ .0052 | <b>.9722 <math>\pm</math> .0099</b> | .9426 $\pm$ .0025                   |
| Isolet            | .8058 $\pm$ .0097 | .9172 $\pm$ .0102 | .9322 $\pm$ .0112                   | .7592 $\pm$ .0087 | .9032 $\pm$ .0142                   | <b>.9422 <math>\pm</math> .0074</b> |
| Ozone             | .9546 $\pm$ .0074 | .9624 $\pm$ .0078 | .9726 $\pm$ .0048                   | .9672 $\pm$ .0075 | <b>.9886 <math>\pm</math> .0098</b> | .9782 $\pm$ .0076                   |
| Parkinsons        | .5652 $\pm$ .0084 | .6526 $\pm$ .0074 | .8017 $\pm$ .0072                   | .6522 $\pm$ .0097 | .5125 $\pm$ .0121                   | <b>.8533 <math>\pm</math> .0074</b> |
| Sonar             | .6012 $\pm$ .0099 | .5912 $\pm$ .0083 | .7723 $\pm$ .0065                   | .5443 $\pm$ .0045 | .5712 $\pm$ .0154                   | <b>.753 <math>\pm</math> .0084</b>  |
| Zoo               | .8232 $\pm$ .0042 | .9642 $\pm$ .0051 | .9776 $\pm$ .0052                   | .9752 $\pm$ .0044 | .8122 $\pm$ .0087                   | <b>.9972 <math>\pm</math> .0074</b> |

**Table 2.** Average test classification accuracy on real-world datasets, including as 4 face datasets and 4 Alzheimer disease (AD) datasets. Experiment results are the average of 20 independent prediction accuracy.

|       | LDA               | LFDA              | LMNN                                | LSDA              | DeepLDA           | DADP                                |
|-------|-------------------|-------------------|-------------------------------------|-------------------|-------------------|-------------------------------------|
| AR    | .8212 $\pm$ .0086 | .9156 $\pm$ .0075 | <b>.9844 <math>\pm</math> .0084</b> | .9786 $\pm$ .0059 | .8556 $\pm$ .0075 | .9836 $\pm$ .0072                   |
| Feret | .4032 $\pm$ .0074 | .5156 $\pm$ .0085 | .5032 $\pm$ .0102                   | .7175 $\pm$ .0174 | .6223 $\pm$ .0082 | <b>.7632 <math>\pm</math> .0075</b> |
| Orl   | .8734 $\pm$ .0062 | .9250 $\pm$ .0514 | .9652 $\pm$ .0047                   | .9525 $\pm$ .0322 | .8172 $\pm$ .0122 | <b>.9752 <math>\pm</math> .0057</b> |
| MRI3  | .4136 $\pm$ .0453 | .3556 $\pm$ .0268 | .5524 $\pm$ .1323                   | .3278 $\pm$ .0351 | .4524 $\pm$ .1422 | <b>.6012 <math>\pm</math> .0072</b> |
| MRI4  | .2732 $\pm$ .1023 | .2950 $\pm$ .0471 | .4286 $\pm$ .0105                   | .3200 $\pm$ .0949 | .4323 $\pm$ .1182 | <b>.4332 <math>\pm</math> .0961</b> |
| PET3  | .3130 $\pm$ .0962 | .3501 $\pm$ .0982 | .5952 $\pm$ .0101                   | .3389 $\pm$ .0315 | .6172 $\pm$ .1022 | <b>.6207 <math>\pm</math> .0418</b> |
| PET4  | .2621 $\pm$ .1049 | .2905 $\pm$ .1055 | <b>.5238 <math>\pm</math> .0840</b> | .2850 $\pm$ .0818 | .3448 $\pm$ .0977 | .4538 $\pm$ .0844                   |

#### 4.1 Experiments on UCI Datasets

We carried out experiments on UCI benchmark datasets, including Breast cancer, Diabetes, Isolet, Zoo etc. The goal of our experiment is to validate the ability of our method to learn a separable subspace for high-dimensional data with small



number of instances. In our experiments, except for DADP, we empirically fine-tuned the projection dimensionality  $p$  and the size of nearest neighborhood  $k$  of KNN classifier. Concretely, we initialized the number of neighbors to  $k' = 7$  for LFDA, LMNN and the size was adjusted to maximize the classification accuracy by 10-fold cross-validation. In addition, for LMNN, we split 20% of training set as validation, other settings like the maximal iteration times, trade-off parameter between loss and regularizer have been fine-tuned according to the accuracy of each experiment. For LSDA, that needs a graph based embedding learning method, we utilized the algorithm LGE and the required affinity graph matrix and constraint graph matrix are constructed in the same way, where KNN with Euclidean distance was used and the neighborhood size was set to 5, weights between two nodes were calculated in HeatKernel mode. As for DeepLDA, we manually altered network architecture for various UCI datasets and we employed a relatively shadow structure here in order to avoid the overfitting problem. The number of the maximized eigenvalues was set  $C - 1$ , where  $C$  is the number of classes. Last, for our method DADP, we gave a rough setting for  $\lambda_1 = 0.01$  and  $\lambda_2 = 0.1$  respectively and fine-tuned the regularization term with a much smaller step to assign a more appropriate weights for all data pairs.

As can be observed from the Table 1, DADP is capable of achieving the highest accuracy compared with other methods or slightly lower than the best accuracy, which demonstrates the effectiveness of multiple-layer nonlinear combinations of features. For UCI datasets, DeepLDA could not perform well in most cases and even performed worst compared with other methods for some datasets. Conclusively, deep neural network based discriminant analysis methods are not suitable for small-scale datasets. Note that, average accuracy of LMNN and LFDA demonstrates the significance of preserving local structure. Lastly, what we want to emphasize is that for Diabetes dataset, the prediction accuracy of most algorithms are below 80% while DADP achieved the accuracy 85.72%. Similarly, for Blood transfusion, DADP achieves the best classification accuracy at least 15% higher than other methods.

## 4.2 Experiments on Several Real-World Datasets

The objective of this experiment is to evaluate the capacity of DADP to handle real-world datasets, including three kinds of face datasets and Alzheimer’s disease dataset.

**Face Datasets.** Face datasets include AR, FERET and Orl databases. AR database contains 3120 grayscale images from 120 persons, 26 images for each person. Images from the same class correspond to one person’s distinct facial expressions pictured in different light condition. Similarly, FERET and Orl databases include  $200 \times 7$ ,  $40 \times 10$  grayscale images respectively. Additionally, for AR and FERET datasets, we pre-processed original images using PCA and projected images on 200-dimensional feature vectors, for Orl datasets, we cropped and resized the original images to  $32 \times 32$  and flatten the pixel matrix to a

1024-dimensional feature vector. Besides, we normalized images by scaling pixel values to  $[0 \sim 1]$ .

As can be seen from Table 2, DADP is able to find a discriminant subspace which preserves separability of different persons. From the perspective of the size of different face datasets, DADP is adaptive to different scaling of datasets. Meanwhile, as expected, DeepLDA still can not perform well owing to the small sample size. Only few hundreds or thousands of pictures are far from sufficient for training deep model in end-to-end fashion.

**Alzheimer’s Disease Datasets.** AD datasets contain 4 types of datasets on Alzheimer’s disease, including MRI3, PET3, MRI4, PET4. Each dataset comprises 202 samples that can be categorized into Alzheimer disease patients (AD), Mild Cognitive Impairment (MCI) and NormalControl (NC). In addition, for MCI, patients can be further divided into MCI-C (those MCI patients convert to AD) and MCI-NC (those MCI patients do not convert to AD). Among them, MRI3 represents samples with 3 classes, MRI4 indicates persons with 4 classes, and similar case for PET datasets. Each row is a 93-dimensional features corresponding to different brain regions extracted from MRI images or PET images.

As can be seen from Table 2, DADP is more suitable for classifying AD samples with three classes comparing with other methods. Especially for MRI3, accuracy of DADP is significantly better than other methods. The prediction results of LFDA and LSDA are around the accuracy of random guess, which illustrates that both two kinds of methods are unable to discriminate AD datasets. For all types of AD datasets, The results of LMNN do not make a big difference and the accuracy of LMNN is slightly lower that of DADP for the first three AD diseases and the highest for PET4, which illustrates that nearest neighbors large margin assumption is appropriate for AD datasets to some extent.

## 5 Conclusion

This paper presents the Depth-Adaptive Discriminant Projection (DADP), a nonlinear discriminant subspace learning model. By virtue of the proposed Adaptive Depth Determination Criterion (ADDC), DADP can adaptively determine the projection dimensionality of each nonlinear layer and the number of nonlinear transformation, which makes DADP more flexible than existing discriminant analysis models combined with Deep Neural Network (DNN). Based on the framework of regularized optimal transport (OT), each layer of DADP learns new weights for each pairwise data, leading to the selection of an optimal projection matrix. Unlike most methods in which weighting scheme is designed under some prior assumption (neighborhood, margin), DADP can flexibly balance between the global and local correlations at a category level by adjusting regularization term of Optimal Transport. Experimental results show that DADP is able to efficiently seek a discriminative subspace for small-scale and high-dimensional datasets.

Similar research, such as manifold learning, component analysis and correlation analysis, can be explored in later work. Future work will consider an objective function for bias vector that has a closed solution for the purpose of further reducing the computation cost and enhancing the adaptivity for small-scale dataset with more complex class distribution.

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